



△ **Marquise-cut** bytownite gemstone

Bytownite

Bytownite is the rarest member of the plagioclase feldspar group; the other members of the group that have gem varieties include labradorite, albite, and oligoclase. Bytownite is seldom found in well-developed crystals, but these can be gemmy when found. Its gemstones are usually faceted, with the transparent gems varying in color from a pale, straw yellow to a light brown. A variety from Mexico is marketed under the name Golden Sunstone, but is different from the various other feldspar sunstones.

Specification

Chemical name Sodium, calcium aluminosilicate | **Formula** $\text{NaAlSi}_3\text{O}_8 - \text{CaAl}_2\text{Si}_2\text{O}_8$ | **Colours** White, grey, yellow, brown
Structure Triclinic | **Hardness** 6–6.5 | **SG** 2.7 | **RI** 1.56–1.57
Lustre Vitreous to pearly | **Streak** White | **Locations** Mexico, Scotland, Greenland, USA, Canada



Small gemmy areas

Bytownite in rock groundmass | Rough | Bytownite seldom forms distinct crystals, but is more often found intergrown with other plagioclases, as here.



Striations

Plagioclase | Rough | This plagioclase specimen shows surface striations, which are the prime characteristic of all plagioclases, including bytownite.



Basalt

Bytownite

Polished bytownite | Cut | This tumble-polished specimen of bytownite in basalt is one of the more unusual occurrences, with the local name of “Lakelandite”.



Pavilion facets visible through table

Fine bytownite | Cut | The bytownite material comprising this stone is unusually flawless, and has been faceted in a step-cut cushion.

Bytownite specimen | Rough | This rough bytownite crystal originates from Ottawa, Canada, and exhibits a remarkable level of clarity and transparency.



Transparent finish



Table facet

Step-cut stone | Cut | The faceter of this emerald-cut bytownite has found an unusually long piece of rough to work from, yielding a striking gem.

Bytownite is one of the minerals known to occur in stony meteorites